A-A-53780
18 April 1990

COMMERCIAL ITEM DESCRIPTION

CATHETER AND NEEDLE UNIT, INTRAVENOUS, Disposable, 50s

The General Services Administration has authorized the use of this commercial item description as a replacement for Military Specification MIL-C-37366A, which is cancelled.

This commercial item description covers radiopaque and non-radiopaque catheter and needle units, with and without syringe, suitable for administering intravenous medication from a single venipuncture.

Salient characteristics:

General. Catheter and needle units shall be furnished as a combination of the following designations:

Catheter (Gage)	Needle Size (Gage)	Catheter length (free inch)	Syringe Class	Composition
A 14	a 16	I 5/8 Inch	0 (no syringe)	1 radiopaque (striped or solid)
B 16	b 17	II 3/4	1 (1 cc syringe)	
C 18	c 18	III 1	2 (2-1/2 or 3 syringe)	2 not radiopaque
D 20	d 19	IV 1-1/8		
E 22	e 20	V 1-1/4		
F 24	f 21	VI 1-1/2		
	g 22	VII 1-5/8		
	h 23	VIII 1-3/4		
	i 24	IX 2		
	j 25	X 2-1/4		
	k 26	XI 2-1/2		

Material.

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Catheter. The catheter shall be fabricated from pliant fluorinated ethylene propylene (FEP) or polypropylene which meets the requirements of the U.S.P. for Class IV plastics. (See Note 2). When the catheter is of two-piece construction the hub shall be fabricated from rigid or semi-rigid plastic which meets the requirements of USP for class II plastics. A polytetrafluroethylene (Teflon) catheter cannula is unacceptable.

Needle. The needle shall be fabricated from type 304 stainless steel in accordance with ASTM Designation F899. The needle hub shall be fabricated either from Class 3 Type 304 steel, FEP, polypropylene, or other plastic meeting USP Class II requirements.

Syringe plunger tip. The syringe plunger tip shall be biologically tested in accordance with either USP Class IV plastics or USP Elastomeric Closures for Injection.

Style and design. The unit shall consist of an outer catheter assembled over an inner hypodermic needle of short bevel design. The needle cannula shall make a safe, sliding fit within the catheter. The proximal end of the catheter shall be permanently secured to a Luer-Lock catheter hub of either one or two piece construction, and in accordance with ANSI/HIMA MD70.1. The proximal end of the needle cannula shall be permanently secured to the distal end of the needle hub. The proximal end of the needle hub shall incorporate a flash chamber and terminate in a female Luer-Slip or Luer-Lock configuration in accordance with ANSI/HIMA MD70.1. A vented plug shall be included. The catheter hub shall be firmly anchored flush against the needle hub to prevent needle movement within the catheter until ready for use. The distal tip of the catheter shall have a smooth taper or step and gentle transition to ease catheter insertion and reduce vascular trauma after venipucture. The needle point tip shall be ground to maximum keeness and the needle point heel shall be dull to prevent coring. Composition 1 catheter and needle units shall be radiopaque in accordance with ASTM Designation F640.

The radiopaque material may be in corporated into the plastic either as stripes or throughout the whole catheter cannula. All units shall be supplied with a needle guard.

The catheter and needle unit shall be in accordance with Defense Personnel Support Center (DPSC) Drawing No. 24380.

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Catheter flexibility. With the needle removed, the catheter shall be wrapped 90 degrees around a 1/2 inch diameter rod without kinking or cracking. Water shall then be admitted through the catheter hub and observed for restricted flow from the distal tip.

Catheter security of assembly. The catheter shall not tear or separate from the catheter hub when the catheter is subjected to the following minimum tensile forces:

GAGE	LOAD (pounds)
14	2.5
16	2.5
18	2.0
20	1.9
22	1.2
24	0.7

Catheter leakage. With the needle removed, the distal tip of the catheter shall be sealed and 20 psig water admitted through the catheter hub. The junction of the catheter to catheter hub shall not leak when observed for 2 minutes.

Heavy metals extractability. Plastic and rubber components coming in contact with intravenous solution or patient's blood shall be tested. The quantity of lead and cadmimum shall not exceed 5 parts per million (ppm) and the amount of barium shall not exceed 10 ppm when tested in accordance with the USP (See Note 3).

Toxicity, pyrogenicity, sterility. The finished catheter and needle unit (and syringe when specified) shall be nontoxic and nonpyrogenic in accordance with the USP for Transfusion and Infusion assemblies (the rabbit pyrogen test) or the Bacterial Endotoxins test (LAL) with the rabbit pyrogen test used for LAL test failure. The catheter and needle unit shall be sterile with a Sterility Assurance Level of 1×10^{-6} in accordance with AAMI guidelines.

Sterility lot and traceability. A sterility lot is that quantity of product sterilized in the same sterilizing chamber at the same time. Each quantity of product shall be traceable to a given manufacturing operation or lot. Each unit package shall be traceable to a given sterility or manufacturing lot (See Note 4).

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Workmanship. The catheter and needle unit (with syringe when specified) shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Package (PG). One package containing 50 individually packaged catheter and needle units, with syringe when specified, and of one size and style only along with instructions, as specified, constitutes one unit. The individual package shall be a sealed, peel-open type container capable of maintaining sterility of contents unless package is damaged or opened.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

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Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, Inspection Level S-1 and AQL 1.0. Test samples shall be selected from each manufacturing lot such as that each material lot is represented in the samples. The unit of product for test purposes shall be one catheter and needle unit (with syringe when specified).

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTES:

(1) Ordering Data. Purchases should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet which shall include this specification and number, and the slash number.

A-A-53780/1	Catheter and Needle Unit, 14 Gage Catheter
A-A-53780/2	Catheter and Needle Unit, 16 Gage Catheter
A-A-53780/3	Catheter and Needle Unit, 18 Gage Catheter
A-A-53780/4	Catheter and Needle Unit, 20 Gage Catheter
A-A-53780/5	Catheter and Needle Unit, 22 Gage Catheter
A-A-53780/6	Catheter and Needle Unit, 24 Gage Catheter

NOTE 2:

The following is acceptable as an alternate to USP Class IV plastics lot testing: The catheter (and syringe when specified) shall be initially qualified for Class IV plastics in accordance with the USP, and simultaneously, tested for tissue culture cytotoxicity in accordance with HIMA "Guideline For Evaluating The Safety of Materials Used in Medical Devices". Subsequent production lots which show greater toxicity than the initial qualification lot shall be subjected to USP Class IV biological testing. In the event of a change in material formulation, USP Class IV initial qualification testing shall again be required.

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NOTE 3:

Certificates of heavy metals testing. A certificate of compliance, stipulating that offered items meet the heavy metals extractable testing, shall be acceptable, provided that the contractor has supplied acceptable items to the Government within the past three years and that there have been no changes to material formulation and method of processing since actual testing was performed.

NOTE 4:

In lieu of having each material batch represented in the finished goods chemical, safety and biological test samples, The following shall be acceptable as an alternate: New raw materials shall be initially qualified through chemical, safety, and biological testing and again testing after assembly, sterilization, and aging. Subsequent incoming materials will be tested physically for material consistency. Any changes to material or processing shall be requalification tested.

Ordering data (Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4610

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